

Tema 4

Mecánica

Hamiltoniano

# Introducción

La mecánica hamiltoniana es el puente entre la mecánica clásica y cuántica

## Transformación de Legendre

Transformación de Legendre: Nos permite aplicar el lagrangiano y obtener el hamiltoniano.

$$\mathcal{L} = \mathcal{L}(q, \dot{q}) \rightarrow \int^* \left( \frac{\partial \mathcal{L}}{\partial \dot{q}} \right) = \overset{P}{\frac{\partial \mathcal{L}}{\partial \dot{q}}} \dot{q} - \mathcal{L} = P\dot{q} - \mathcal{L} = \mathcal{H}$$

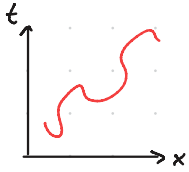
Aplicación de transformada de Legendre al lagrangiano

Ya habíamos visto que si el Lagrangiano no depende de  $t$ , el hamiltoniano es una cantidad conservada y es la energía total del sistema.

$$\mathcal{H} = p\dot{q} - \mathcal{L} = m\dot{q}\dot{q} - \mathcal{L} = 2T - T + U = T + U \rightarrow \text{si las coordenadas naturales}$$

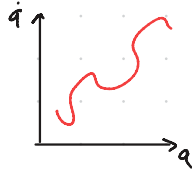
Coordenadas naturales  $\rightarrow$  Sistema Inercial  $\rightarrow$  No hay rotación

## Espacio de Fases



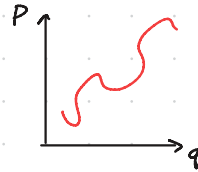
Newtoniano:

- Espacio Físico



Lagrangiano:

- Espacio de Estados



Hamiltoniano:

- Espacio de fases

## Ecuaciones de Hamilton

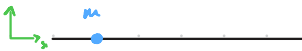
Por cada grado de libertad  $\rightarrow$  Obtenemos un par de EDOs de primer orden

$$\dot{q}_i = \frac{\partial \mathcal{H}}{\partial p_i}$$

$$\dot{p}_i = - \frac{\partial \mathcal{H}}{\partial q_i}$$

### Ejemplo 1

$$U = U(x) \quad 1 \text{ gde}(x) \quad \text{Ligaduras} = 0$$



$$T = \frac{1}{2} m \dot{x}^2$$

$$U = U(x)$$

$$\mathcal{L} = T - U = \frac{1}{2} m \dot{x}^2 - U(x)$$

$$\mathcal{H} = T + U = \frac{1}{2} m \dot{x}^2 + U(x)$$

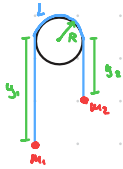
$$\text{Ec. Euler-Lagrange: } \frac{\partial \mathcal{L}}{\partial x} = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} \rightarrow m \ddot{x} = \frac{\partial U}{\partial x}$$

$$\text{Ec. Hamilton: } 1 \text{ gde} \rightarrow 2 \text{ ec.}$$

$$\dot{x} = \frac{\partial \mathcal{H}}{\partial p} = \frac{p}{m} \rightarrow p = m \dot{x} \rightarrow \text{Momento lineal}$$

$$\dot{p} = - \frac{\partial \mathcal{H}}{\partial x} \rightarrow \dot{p} = - \frac{\partial U}{\partial x} \rightarrow 2^\circ \text{ Ley Newton}$$

## Ejemplo 2



Ligadura:  $L = y_1 + y_2 + nR$ ,  $y_1 = y_2$

(coordenadas  $(y_1, y_2)$ )

$$T = \frac{1}{2} m_1 \dot{y}_1^2 + \frac{1}{2} m_2 \dot{y}_2^2 \quad U = -m_1 g y_1 - m_2 g y_2$$

Aplico ligadura:  $y_2 = L - y_1 - nR$

$$T = \frac{1}{2} (m_1 + m_2) \dot{y}_1^2 \quad U = -m_1 g y_1 - m_2 g (L - nR - y_1) = (m_2 - m_1) y_1 - m_2 (L - nR)$$

$$\mathcal{L} = \frac{1}{2} (m_1 + m_2) \dot{y}_1^2 - (m_2 - m_1) g y_1 + m_2 g (L - nR)$$

Yo no se si mis coordenadas son naturales:  $\mathcal{H} = P_1 \dot{q}_1 + P_2 \dot{q}_2 - \mathcal{L}$

$\mathcal{H} = T + U = \frac{1}{2} (m_1 + m_2) \dot{y}_1^2 + (m_2 - m_1) g y_1 - m_2 g (L - nR) \rightarrow$  Pero tenemos que quitar  $y_1$  por el Hamiltoniano en función de  $p_1$ .

$$P_1 = \frac{\partial \mathcal{L}}{\partial \dot{y}_1} = (m_1 + m_2) \dot{y}_1 \rightarrow \dot{y}_1 = \frac{P_1}{m_1 + m_2} \rightarrow \text{Sustituyo el hamiltoniano}$$

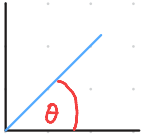
$$\mathcal{H} = \frac{1}{2} (m_1 + m_2) \frac{P_1^2}{(m_1 + m_2)^2} + (m_2 - m_1) g y_1 - m_2 g (L - nR) = \frac{P_1^2}{2(m_1 + m_2)} + (m_2 - m_1) g y_1 + m_2 g (L - nR)$$

Es Hamilton:

$$\dot{y}_1 = \frac{\partial \mathcal{H}}{\partial P_1} = \frac{P_1}{m_1 + m_2} \quad \dot{P}_1 = -\frac{\partial \mathcal{H}}{\partial y_1} = (m_2 - m_1) g$$



### Ejemplo 3



$$\begin{aligned} T &= \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) \\ U &= U(r) \end{aligned} \quad \left\{ \begin{aligned} \mathcal{L} &= \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) - U(r) \\ \mathcal{H} &= \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) + U(r) \end{aligned} \right.$$

$$p_r = \frac{\partial \mathcal{L}}{\partial \dot{r}} = m \dot{r} \Rightarrow \dot{r} = \frac{p_r}{m}$$

$$p_\theta = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m r^2 \dot{\theta} \rightarrow \dot{\theta} = \frac{p_\theta}{m r^2}$$

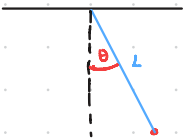
$$\mathcal{H} = \frac{1}{2} m \left( \frac{p_r^2}{m^2} + \frac{p_\theta^2}{m^2 r^2} \right) + U(r)$$

Ec. Hamilton: 4 ecs

$$r \begin{cases} \dot{r} = \frac{\partial \mathcal{H}}{\partial p_r} = m \frac{p_r}{m^2} = \frac{p_r}{m} \rightarrow p_r = m \dot{r} \\ \dot{p}_r = -\frac{\partial \mathcal{H}}{\partial r} = -\frac{p_\theta^2}{m r^3} - \frac{\partial U}{\partial r} \end{cases}$$

$$\theta \begin{cases} \dot{\theta} = \frac{\partial \mathcal{H}}{\partial p_\theta} = \frac{p_\theta}{m r^2} \Rightarrow p_\theta = m r^2 \dot{\theta} \\ \dot{p}_\theta = -\frac{\partial \mathcal{H}}{\partial \theta} = 0 \rightarrow p_\theta = \text{cte} \end{cases}$$

### Ejemplo 4



Ligaduras:  $r = L$

Polares  $(r, \theta)$

$$\begin{aligned} T &= \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) = \frac{1}{2} m L^2 \dot{\theta}^2 \\ U &= -mgL \cos \theta \end{aligned} \quad \left\{ \begin{aligned} \mathcal{L} &= \frac{1}{2} m L^2 \dot{\theta}^2 + mgL \cos \theta \\ \mathcal{H} &= \frac{1}{2} m L^2 \dot{\theta}^2 - mgL \cos \theta \end{aligned} \right.$$

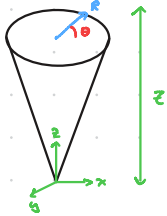
$$p_\theta = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m L^2 \dot{\theta} \rightarrow \dot{\theta} = \frac{p_\theta}{m L^2}$$

$$\mathcal{H} = m L^2 \frac{p_\theta^2}{m^2 L^4} - mgL \cos \theta = \frac{p_\theta^2}{2 m L^2} - mgL \cos \theta$$

$$\dot{\theta} = \frac{\partial \mathcal{X}}{\partial P_{\theta}} = \frac{P_{\theta}}{mL^2} = \left[ mgl \sin \theta + C_1 \right] \frac{1}{mL^2} = \frac{g}{L} \sin \theta + C_1'$$

$$P_{\theta} = \frac{\partial \mathcal{X}}{\partial \theta} = mgl \sin \theta \rightarrow P_{\theta} = \int mgl \sin \theta d\theta = mgl \cos \theta + C_1$$

## Ejemplo 5



Coordenadas Cilíndricas  $(r, \theta, z)$

Ligadura:  $r = cz \rightarrow \dot{r} = c \dot{z}$

$$T = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + \dot{z}^2) = \frac{1}{2} m (c^2 \dot{z}^2 + c^2 \dot{\theta}^2 + \dot{z}^2) = \frac{1}{2} m ((c^2+1) \dot{z}^2 + c^2 \dot{\theta}^2)$$

$$U = mgz$$

$$\mathcal{L} = \frac{1}{2} m ((c^2+1) \dot{z}^2 + c^2 \dot{\theta}^2) - mgz$$

$$\mathcal{H} = \frac{1}{2} m ((c^2+1) \dot{z}^2 + c^2 \dot{\theta}^2) + mgz \rightarrow \text{Asumiendo coordenadas naturales}$$

Saco los momentos conjugados de  $\theta$  y  $z$ .

$$P_{\theta} = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = mc^2 \dot{\theta} \rightarrow \dot{\theta} = \frac{P_{\theta}}{mc^2}$$

$$P_z = \frac{\partial \mathcal{L}}{\partial \dot{z}} = m(c^2+1) \dot{z} \rightarrow \dot{z} = \frac{P_z}{m(c^2+1)}$$

Sustituimos:

$$\mathcal{H} = \frac{1}{2} m \left( (c^2+1) \frac{P_z^2}{m^2(c^2+1)^2} + c^2 \frac{P_{\theta}^2}{m^2 c^4} \right) + mgz$$

$$\mathcal{H} = \frac{1}{2} \left( \frac{P_z^2}{m(c^2+1)} + \frac{P_{\theta}^2}{mc^2} \right) + mgz$$

Sino se si coordenadas naturales:

$$\mathcal{H} = \sum P_i \dot{q}_i - \mathcal{L} = P_{\theta} \dot{\theta} + P_z \dot{z} - \mathcal{L} = P_{\theta} \frac{P_{\theta}}{mc^2} + P_z \frac{P_z}{m(c^2+1)} - \frac{1}{2} \left( \frac{P_{\theta}^2}{mc^2} + \frac{P_z^2}{m(c^2+1)} \right) + mgz$$

$$\mathcal{H} = \frac{1}{2} \left( \frac{P_z^2}{m(c^2+1)} + \frac{P_{\theta}^2}{mc^2} \right) + mgz$$

Ex. Hamilton: 2 qde  $\rightarrow$  4 ecuaciones

$$\dot{z} = \frac{\partial \mathcal{H}}{\partial z} = \frac{P_z}{m(c^2+1)}$$

$$\dot{\theta} = \frac{\partial \mathcal{H}}{\partial \theta} = \frac{P_\theta}{m c^2 \dot{z}} \rightarrow P_\theta = m c^2 \dot{z} \dot{\theta} \rightarrow \text{Momento angular}$$

$$\dot{P}_z = -\frac{\partial \mathcal{H}}{\partial z} = \frac{P_\theta^2}{m c^2 \dot{z}^3} - m g$$

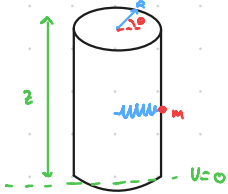
$$\dot{P}_\theta = -\frac{\partial \mathcal{H}}{\partial \theta} = 0 \rightarrow \text{Conservaci3n Momento Angular}$$

## Coordenadas Ignorables

Mecanica Lagrangiana:  $\frac{\partial \mathcal{L}}{\partial q_i} = 0 \rightarrow$  Coordenada ciclica

Analogamente:  $\dot{P}_i = -\frac{\partial \mathcal{H}}{\partial q_i} = 0 \rightarrow$  Coordenada Ignorable

## Ejemplo 6



Ligadura:  $r = R$   $\dot{r} = 0$  Coordenadas Cilindricas  $(r, \theta, z)$

$$T = \frac{1}{2} m (\dot{z}^2 + r^2 \dot{\theta}^2 + \dot{z}^2) = \frac{1}{2} m (R^2 \dot{\theta}^2 + \dot{z}^2)$$

$$U = U_{\text{grav}} + U_{\text{muelle}} = m g z + \frac{1}{2} k x^2 \rightarrow U = m g z + \frac{1}{2} k (z^2 + R^2)$$

$$\mathcal{L} = \frac{1}{2} m (R^2 \dot{\theta}^2 + \dot{z}^2) - m g z - \frac{1}{2} k (z^2 + R^2)$$

$$\mathcal{H} = \sum P_i \dot{q}_i - \mathcal{L} = P_\theta \dot{\theta} + P_z \dot{z} - \mathcal{L} =$$

Encontramos momentos conjuntos:

$$P_\theta = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m R^2 \dot{\theta} \rightarrow \dot{\theta} = \frac{P_\theta}{m R^2}$$

$$P_z = \frac{\partial \mathcal{L}}{\partial \dot{z}} = m \dot{z} \rightarrow \dot{z} = \frac{P_z}{m}$$

$$\mathcal{H} = P_\theta \frac{P_\theta}{m R^2} + P_z \frac{P_z}{m} - \frac{1}{2} m \left( R^2 \frac{P_\theta^2}{m^2 R^2} + \frac{P_z^2}{m^2} \right) + m g z + \frac{1}{2} k (z^2 + R^2)$$

$$\mathcal{H} = \frac{1}{2} \left( \frac{P_\theta^2}{mR^2} + \frac{P_z^2}{m} \right) + mgz + \frac{1}{2} k (z^2 + R^2)$$

Ej. Hamilton:

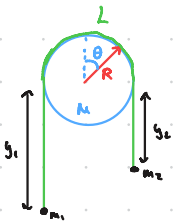
$$\dot{\theta} = \frac{\partial \mathcal{H}}{\partial P_\theta} = \frac{P_\theta}{mR^2}$$

$$\dot{z} = \frac{\partial \mathcal{H}}{\partial P_z} = \frac{P_z}{m}$$

$$\dot{P}_\theta = -\frac{\partial \mathcal{H}}{\partial \theta} = 0$$

$$\dot{P}_z = -\frac{\partial \mathcal{H}}{\partial z} = -mg + k_z$$

## Ejemplo 7



Cilindro:  $I = \frac{1}{2} MR^2$

ligaduras:  $y_1 = y_2$ ,  $y_1 + y_2 + vR = L$ ,  $\dot{y} = R\dot{\theta}$ ,  $y = R\theta$

$$T = \frac{1}{2} m_1 \dot{y}_1 + \frac{1}{2} m_2 \dot{y}_2 + \frac{1}{2} I \omega^2 = \frac{1}{2} (m_1 + m_2) \dot{y}_2 + \frac{1}{4} MR^2 \dot{\theta}^2 = \frac{1}{2} (m_1 + m_2) \dot{y}_2^2 + \frac{1}{2} MR^2 \frac{\dot{y}_2^2}{R^2}$$

$$T = \frac{1}{2} (m_1 + m_2 + \frac{M}{2}) \dot{y}_2^2$$

$$V = -m_1 g y_1 - m_2 g y_2 = -m_1 g (L - y_2 - vR) - m_2 g y_2 = (m_1 - m_2) g y_2 + m_1 g vR$$

$$\mathcal{L} = \frac{1}{2} (m_1 + m_2 + \frac{M}{2}) \dot{y}_2^2 - (m_1 - m_2) g y_2 - m_1 g vR$$

$$\mathcal{H} = P_2 \dot{y}_2 - \mathcal{L}$$

Encuentro momento asociado:

$$P_2 = \frac{\partial \mathcal{L}}{\partial \dot{y}_2} = (m_1 + m_2 + \frac{M}{2}) \dot{y}_2 \rightarrow \dot{y}_2 = \frac{P_2}{m_1 + m_2 + \frac{M}{2}}$$

$$\mathcal{H} = P_2 \frac{P_2}{m_1 + m_2 + \frac{M}{2}} - \frac{1}{2} (m_1 + m_2 + \frac{M}{2}) \left( \frac{P_2}{m_1 + m_2 + \frac{M}{2}} \right)^2 + (m_1 - m_2) g y_2 - m_1 g vR$$

Ej. Hamilton:  $y_2 = \frac{\partial \mathcal{H}}{\partial P_2} = \frac{P_2}{m_1 + m_2 + \frac{M}{2}}$   $\dot{P}_2 = \frac{\partial \mathcal{H}}{\partial y_2} = (m_1 - m_2) g$

## Ejemplo 8



$$\vec{F} = (-kx)\hat{x} + cy\hat{y}$$

$$U(x,y) = - \int F dr = - \int -kx dx + cy dy \Rightarrow U(x,y) = \frac{kx^2}{2} - cy$$

$$T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2)$$

$$\mathcal{L} = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) - \frac{1}{2} kx^2 + cy$$

$$\mathcal{H} = p_x \dot{x} + p_y \dot{y} - \mathcal{L}$$

Calcular momentos asociados:

$$p_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x} \rightarrow \dot{x} = \frac{p_x}{m}$$

$$p_y = \frac{\partial \mathcal{L}}{\partial \dot{y}} = m\dot{y} \rightarrow \dot{y} = \frac{p_y}{m}$$

$$\mathcal{H} = T + U = \frac{1}{2} m \left( \frac{p_x^2}{m^2} + \frac{p_y^2}{m^2} \right) + \frac{1}{2} kx^2 - cy = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{1}{2} kx^2 - cy$$

Ec. Hamilton:

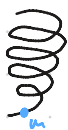
$$\dot{x} = \frac{\partial \mathcal{H}}{\partial p_x} = \frac{p_x}{m}$$

$$\dot{y} = \frac{\partial \mathcal{H}}{\partial p_y} = \frac{p_y}{m}$$

$$\dot{p}_x = -\frac{\partial \mathcal{H}}{\partial x} = -kx$$

$$\dot{p}_y = -\frac{\partial \mathcal{H}}{\partial y} = -c$$

## Ejemplo 9



Cilindricas  $(\rho, \theta, z)$

Ligaduras:  $z = c\theta$   $\rho = R$   
 $\dot{z} = c\dot{\theta}$

$$T = \frac{1}{2} m (\dot{\rho}^2 + \rho^2 \dot{\theta}^2 + \dot{z}^2) = \frac{1}{2} m (R^2 \dot{\theta}^2 + c^2 \dot{\theta}^2) = \frac{1}{2} m (R^2 + c^2) \dot{\theta}^2$$

$$U = mgz = mgc\theta \rightarrow \mathcal{L} = \frac{1}{2} m (R^2 + c^2) \dot{\theta}^2 - mgc\theta$$

$$\mathcal{H} = p_\theta \dot{\theta} - \mathcal{L} = \frac{p_\theta^2}{m(R^2 + c^2)} - \frac{1}{2} m (R^2 + c^2) \frac{p_\theta^2}{m^2(R^2 + c^2)} + mgc\theta = \frac{p_\theta^2}{2m(R^2 + c^2)} + mgc\theta$$

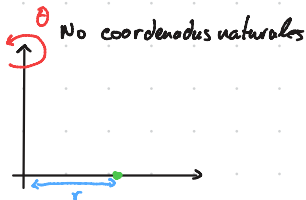
$$p_\theta = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m(R^2 + c^2) \dot{\theta} \rightarrow \dot{\theta} = \frac{p_\theta}{m(R^2 + c^2)}$$

E.c. Hamilton:

$$\dot{\theta} = \frac{\partial \mathcal{H}}{\partial p_\theta} = \frac{p_\theta}{m(R^2 + c^2)} = \frac{-mgct + c_1}{m(R^2 + c^2)}$$

$$p_\theta = \frac{\partial \mathcal{H}}{\partial \dot{\theta}} = -mgc \rightarrow p_\theta = \int -mgc dt = -mgct + c_1$$

## Ejemplo 10



Polares  $(r, \theta)$

Ligadura:  $\dot{\theta} = \omega$

$$T = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) = \frac{1}{2} m (\dot{r}^2 + r^2 \omega^2) = \mathcal{L}$$

$$U = 0$$

$$\mathcal{H} = p_r \dot{r} - \mathcal{L} = \frac{p_r^2}{m} - \frac{1}{2} (\dot{r}^2 + r^2 \omega^2)$$

$$\dot{r} = \frac{\partial \mathcal{L}}{\partial p_r} = m \dot{r} \rightarrow \dot{r} = \frac{p_r}{m}$$

Demostración no naturales:

$$\mathcal{H} = T + U = \frac{1}{2} m \left( \frac{p_r^2}{m^2} + r^2 \omega^2 \right) = \frac{p_r^2}{2m} + \frac{1}{2} m r^2 \omega^2 \rightarrow \text{No es el mismo resultado}$$

## Sistemas Dinámicos y Diagramas de Fase

Los espacios a menudo requieren mas dimensión  $\rightarrow$  Manifold

Propiedades de diagrama de fases:

- 2 orbitas no se pueden cruzar
- El valor del hamiltoniano es cte a lo largo de la curva
- Diferentes curvas = Distinto valor de  $\mathcal{H}$

Geodésica: Camino mas corto entre 2 puntos

Aplicación a la relatividad:

- la geodésica de las particulas en el espacio-tiempo curvo pueden derivarse como curvas de flujo hamiltoniano

## Ejemplo 11

1 gde  $\rightarrow x$

No ligaduras



$$\left. \begin{aligned} T &= \frac{1}{2} m \dot{x}^2 \\ U &= \frac{1}{2} k x^2 \end{aligned} \right\} \mathcal{L} = T - U = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} k x^2$$

$$\mathcal{H} = p_x \dot{x} - \mathcal{L}$$

Encontramos momentos asociados:

$$p_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m \dot{x} \rightarrow \dot{x} = \frac{p_x}{m}$$

$$\mathcal{H} = \frac{p_x^2}{m} - \frac{1}{2} m \frac{p_x^2}{m^2} + \frac{1}{2} k x^2 = \frac{1}{2} \left[ \frac{p_x^2}{m} + k x^2 \right]$$

Ec. Hamilton:

$$\dot{x} = \frac{\partial \mathcal{H}}{\partial p_x} = \frac{p_x}{m}$$

$$\dot{p}_x = -\frac{\partial \mathcal{H}}{\partial x} = -k x$$

$$\text{Orbitas: } \mathcal{H} = \frac{p_x^2}{2m} + \frac{1}{2} k x^2 \rightarrow 1 = \frac{1}{2} \left[ \frac{p_x^2}{2m \mathcal{H}} + \frac{k x^2}{\mathcal{H}} \right] \rightarrow \text{Esto tiene forma de elipse}$$

$$\text{Elipse: } 1 = \frac{x^2}{a^2} + \frac{y^2}{b^2} \rightarrow a^2 = 2m \mathcal{H} \quad b^2 = \frac{2 \mathcal{H}}{k}$$



## Ejemplo 12



$1 \text{ gdl} \rightarrow y$

No ligados

$$\left. \begin{aligned} T &= \frac{1}{2} m \dot{y}^2 \\ U &= -mgy \end{aligned} \right\} \begin{aligned} \mathcal{L} &= \frac{1}{2} m \dot{y}^2 + mgy \\ \mathcal{H} &= \frac{1}{2} m \dot{y}^2 - mgy \end{aligned}$$

$\rightarrow$  Asumimos coordenadas naturales

$$p_y = \frac{\partial \mathcal{L}}{\partial \dot{y}} = m\dot{y} \rightarrow \dot{y} = \frac{p_y}{m}$$

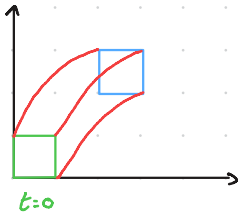
$$\mathcal{H} = \frac{1}{2} m \frac{p_y^2}{m^2} - mgy = \frac{1}{2} \frac{p_y^2}{m} - mgy$$

$\mathcal{L}_c$  Hamilton:

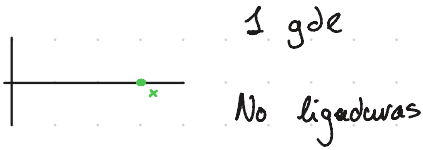
$$\dot{y} = \frac{\partial \mathcal{H}}{\partial p} = \frac{p_y}{m} = \frac{p_0 + mgt}{m} \rightarrow y = \int \left( \frac{p_0}{m} + g \right) dt \Rightarrow y = y_0 + \frac{p_0}{m} t + \frac{1}{2} g t^2$$

$$\dot{p} = -\frac{\partial \mathcal{H}}{\partial y} = -mg \rightarrow p = \int -mg dt = -p_0 - mgt$$

Damos valores a  $t$



### Ejemplo 13



$$F_x = -kx^3 \rightarrow U = -\int F dx = \frac{1}{4} kx^4 \quad T = \frac{1}{2} m\dot{x}^2$$

$$\mathcal{L} = \frac{1}{2} m\dot{x}^2 - \frac{1}{4} kx^4 \quad p = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x} \rightarrow \dot{x} = \frac{p}{m}$$

$$\mathcal{H} = \frac{1}{2} m\dot{x}^2 + \frac{1}{4} kx^4$$

$$\mathcal{H} = \frac{p^2}{2m} + \frac{1}{4} kx^4 \Rightarrow \mathcal{H} = \frac{p^2}{2m} + \frac{k(x')^2}{4} \rightarrow \text{Elipse} \rightarrow a^2 = 2m \cdot ( \quad ) \quad b^2 = \frac{4x}{k}$$